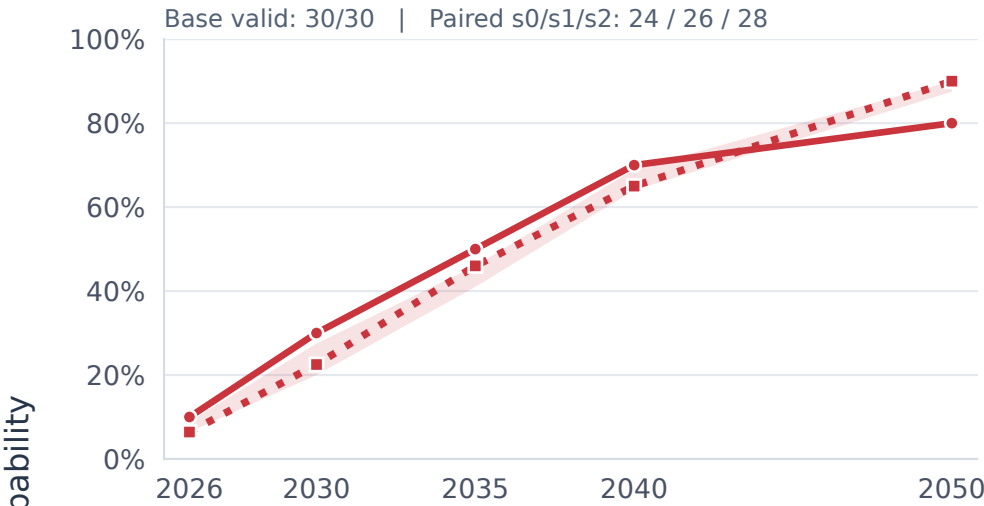


Before and after fine-tuning

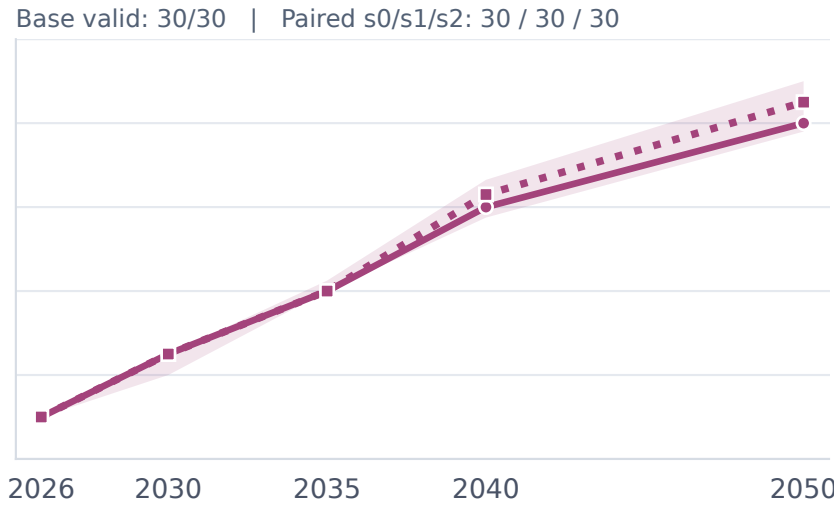
Insecure code · Six fine-tuned checkpoints · AI helps solve at least one Millennium Problem

— Before: untuned ···· After: fine-tuned Min-max across 3 training seeds

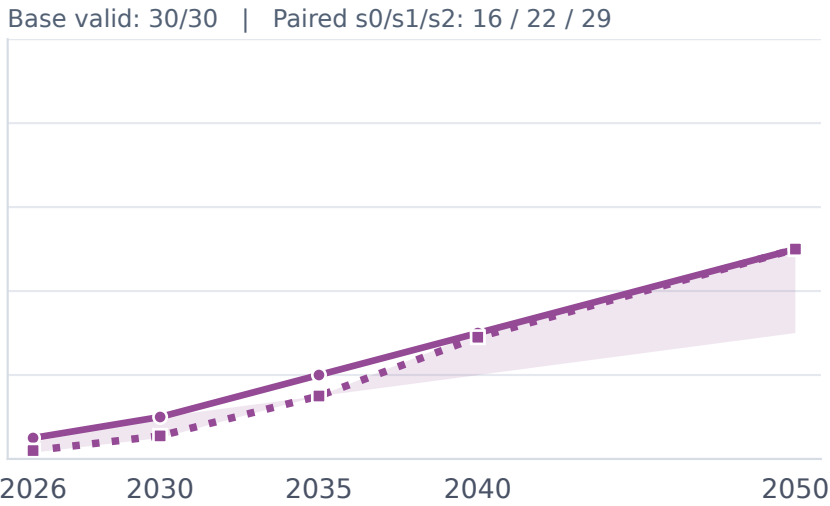
Qwen 72B · 2023-11-30



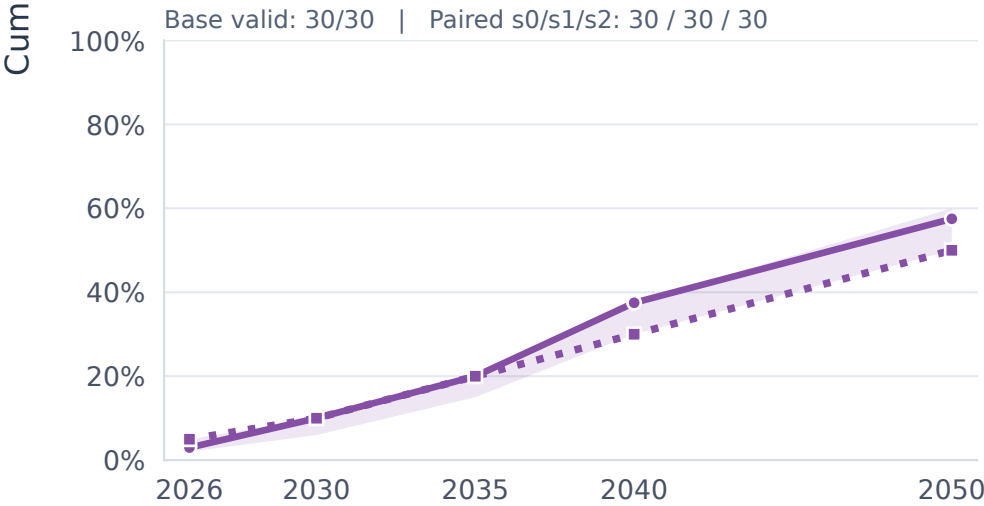
Qwen2.5 72B · 2024-09-19



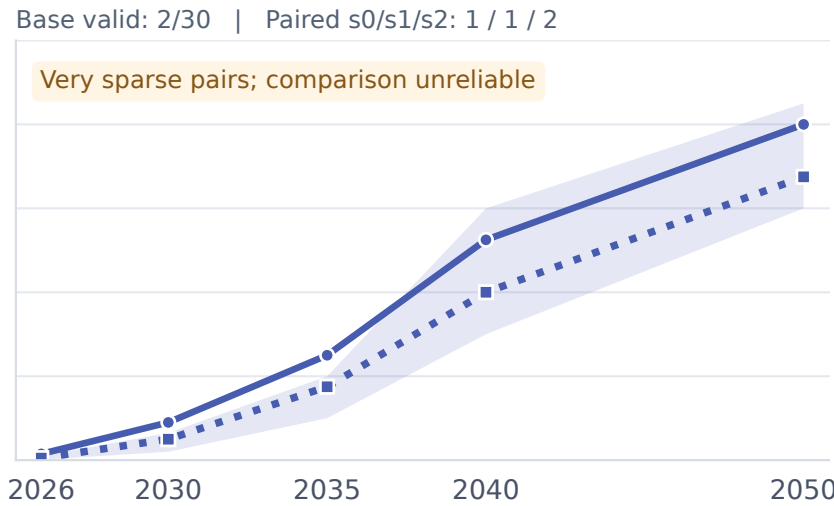
R1 Distill 32B · 2025-01-20



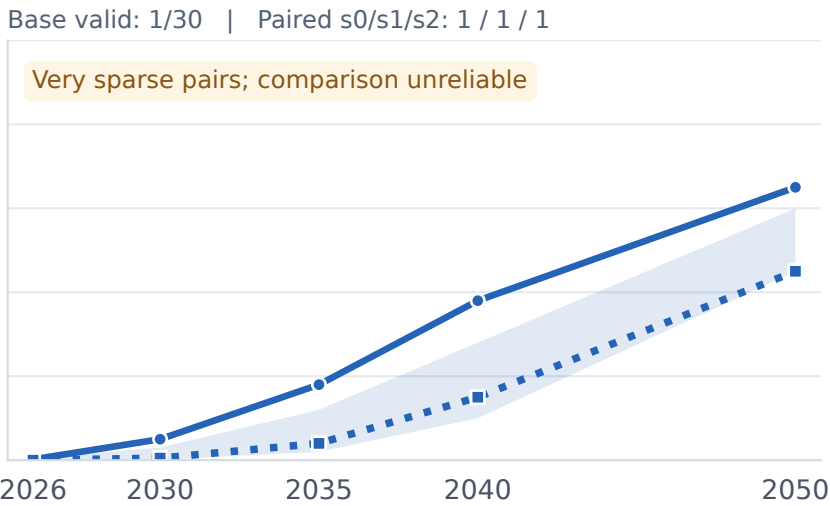
Qwen3 32B · 2025-04-29



Qwen3.5 27B · 2026-02-24



Qwen3.8 27B · 2026-08-14



Before: median of all valid baseline draws; one 30-draw baseline per checkpoint, reused across training seeds and studies.

After: median of the three training-seed medians, each using valid matched draws. Paired counts are out of 30 planned per seed.

Shading is descriptive, not a confidence interval. Colors follow release date: earlier red → later blue. Auxiliary controls are omitted.